

smallcases Investment Performance: A Returns-Based Factor Analysis

Rajan Raju^{*†}

Feb 2023

Abstract

We examine the performance of 36 smallcases between December 2019 and December 2022, comparing their performance against market-based and strategy indices, and four- and six-factor models. The results show evidence of targeted factor exposures for a few smallcases, but for most smallcases, the high idiosyncratic risk of concentrated portfolios overwhelms any potential factor exposure. In general, smallcases in our sample are exposed to the *Small* Size factor, which will likely constrain the capacity. Realised returns are likely to be lower due to fees, slippage, transaction costs, and taxes. Despite excluding these, there is no evidence of alpha in our sample controlling for common academic factors. The findings provide preliminary information for investors and investment professionals who want to better understand the factor exposures of smallcases.

Keywords— Factors, smallcases, Investment Strategy, Indian Equity

JEL Classification Codes G00, G11, C15

^{*}Email: rajanraju@invespar.com, Phone: +65.62380361

[†]Thanks to Mr Anish Teli for his comments on an earlier draft of paper.

1. Introduction

This paper aims to examine the performance of strategies on the Smallcase platform in India. The Indian investment landscape is highly regulated, with the Securities and Exchange Board of India (SEBI) playing a crucial role in ensuring compliance and protecting investors' interests. The Smallcase platform¹, launched in 2016, allows investors to invest in pre-built, diversified portfolios of stocks and ETFs, known as "smallcases". These portfolios are curated by "expert" investors based on various investment themes and strategies. Several of these smallcases rely on systematic, rule-based algorithms that create concentrated portfolios of stocks exposed to the particular characteristic the manager is targeting. As factor strategies are systematic, these are popular on the platform with several smallcases targeting Momentum, Value, or Quality factors. The platform has seen significant growth in recent years, with an estimated ₹29000+ crore transacted as of 2022 and more than 1,000 strategies² available on the platform.

There is little formal research on performance strategies on the Smallcase platform. Given the popularity of factor-style smallcases, this paper seeks to contribute to the academic literature by analysing the performance and estimating factor exposure using a returns-based analysis of a sample of smallcases. The research findings will provide preliminary information for investors and investment professionals looking to understand the smallcases better. Given the newness of the platform, of the thousand smallcases, only a few have a long enough history to allow for a formal return-based analysis. There are some preliminary conclusions from the analysis. Although some smallcases show exposure to targeted factors, for many, the high idiosyncratic risk arising from the concentrated portfolios overwhelms any evidence of factor exposure. With strategies exposed to mid- and small-cap stocks, the idiosyncratic risk is higher.

2. Our sample of smallcases

Of the long list of smallcases available on the Smallcase platform, we searched for strategies with at least 36 consecutive months of available price information as of December 2022. Coincidentally, this resulted in a list of 36 strategies. Smallcase categorises strategies into several categories, including Asset Allocation, Dividend, Factor Investing, Momentum, Quality, and Value. Some small cases are listed in several categories. We categorised the 36 strategies into single style categories that best describe the strategy. As the focus of our paper is to study the factor exposures of smallcases in our sample, our priority is to map the smallcases to mainstream academic factors. The style categories Momentum, Quality, and Value have broad reference to individual academic factors. Although Smallcase does not have a Low Volatility category, one fund in our sample is so obviously focused on low volatility that we have categorised this fund accordingly. Some funds that follow a multifactor investment style in our sample are categorised as MultiFactor. Smallcase has Asset Allocation and Dividend as separate categories which do not correspond to academic factors. So we retain these two as distinct from the more obvious factor-style funds.

¹<https://www.smallcase.com/>

²Taken from smallcase website on January 23, 2023.

Table 1 summarises the strategies in our sample categorised by style. Strategy Type is another categorisation of the strategy by Smallcase. Most strategies are “model-based”: relying on a quantitative model with systematic rebalancing. Managers of such strategies follow the output of their proprietary models, but it is unclear whether these outcomes can be ignored or changed by managers. Smallcase does not make any representations on behalf of the strategies, making adherence to model outputs opaque to investors. Smallcase has other strategy types: All Weather Investing, Thematic, Smart Beta and ETF. These types are not clearly defined on the smallcase website or disclosures.

Rebalancing frequency refers to the frequency of the research team of the strategy “*reviews this smallcase to add or drop stock/ETFs if applicable*”³. This can be Weekly, Monthly, Quarterly, Annual or not disclosed. The actual rebalancing date is not disclosed, but the dates of the most recent “review” and the upcoming review are disclosed in the fact sheets.

Smallcase categorises each strategy by market capitalisation, similar to how the Securities and Exchange Board of India (SEBI) has mandated for all mutual funds. It is important to note that while Smallcase and SEBI use similar terms, the underlying definitions and processes are substantially different. We will briefly cover a few differences. First, the Smallcase market cap categorisation is not calculated “*if the creator/manager of the smallcase specifically defines a particular category*”. If the manager has not defined the category, Smallcase calculates the category based on the sum of the weights of the categories of the constituent stocks⁴. The universe from which stocks can be selected is stated in the fact sheet under a subsection titled “Defining the universe”. In our sample, the majority of smallcases have a wide universe⁵, which according to the SEBI classification schema will require the fund manager to restrict exposure to various categories of market cap. On the other hand, Smallcase computes the market cap category based on the market cap of the constituents of the strategy at a point in time. So, on the review date, if “*the sum of weights of constituent large cap stocks is greater than 50%*”, the particular smallcase will be classified as Largecap. If, on a subsequent date, after rebalancing, the sum of weights of constituent stocks classified as mid-cap is greater than 50%, then the smallcase would be re-categorised as Midcap! Smallcase ranks all stocks in order of market capitalisation and classifies stocks less than 101 as Large Cap; those ranked above 100 and below 251 as Mid Cap. Stocks ranked above 250 are classified as Small Cap. With frequent rebalancing from a wide universe, there is every likelihood for a particular smallcase to change market cap categories, unlike under the SEBI classification process. Unfortunately, the time series of market cap categorisation is not publicly available.

Second, the market cap category is used by Smallcase to assign benchmarks to strategies. The benchmarks

³From smallcase fact sheets.

⁴The disclosures in the fact sheets of each smallcase detail the calculation procedure.

⁵With phrases like “*All publicly traded companies on the National Stock Exchange of India, covering are included in the universe*” and “*textitAll publicly traded companies on the National Stock Exchange of India, covering 90% market capitalization, are included in the universe*”.

Table 1: Summary description and characteristics of smallcase strategies analysed

Style	code	Name	Summary	Strategy Type	Rebal- ance Freq	Manager	Access Type	Market Cap Cat	Risk Cat	Benchmark	Num Hold- ings
Asset Allocation	SCAW_0001	All Weather Investing	One investment for all market conditions. Works for everyone	All Weather Investing	Q	Windmill Capital	FREE	None	Low	NIFTY 100	4
	SCET_0004	Top 100 Stocks	India's most powerful companies in one portfolio. Solid stability	ETF	Q	Windmill Capital	FREE	None	Medium	NIFTY 100	2
	SCET_0005	Equity and Gold	Create wealth with equities, stay protected with Gold. The sweet spot	ETF	Q	Windmill Capital	FREE	None	Low	NIFTY 100	2
	TRLMO_0003	TrendlineZ Balanced	A conservative strategy with dynamic asset allocation between equity and debt.	Model-based	-	TrendlineZ	PAID	None	Low	NIFTY 500	22
Dividends	SCMO_0013	Dividend Stars	High dividend yield companies with a track record of not cutting dividends	Model-based	A	Windmill Capital	PAID	Small cap	Medium	NIFTY 500	11
	SCMO_0014	Dividend Aristocrats	Companies that have been consistently increasing dividends. Extra goodness	Model-based	A	Windmill Capital	FREE	Large cap	Medium	NIFTY 500	12
Low Vol	SCMO_0015	Safe Haven	Low beta stocks that are rated .AaBUY,Äa, to help protect against market volatility	Model-based	Q	Windmill Capital	FREE	Multi cap	Medium	NIFTY 500	10
Momentum	CMMO_0001	Capitalmind Momentum	Up to 25 stocks selected based on sharpe ratios and liquidity. Maximum investable up to 20L.	Model-based	W	Capital-mind	PAID	Large cap	Low	NIFTY 500	25
	FBCMO_0003	Flameback Large Cap Momentum Factor	Top 15 stocks that score high on momentum factor out of a universe of top 100 stocks by market cap.	Model-based	M	Flameback Capital	PAID	None	Medium	NIFTY 100	15
	FBCMO_0004	Flameback MultiCap Momentum Factor	Top 50 stocks that score high on momentum factor out of the universe of top 500 stocks.	Model-based	M	Flameback Capital	PAID	None	Medium	NIFTY 500	50
	LTDMO_0001	Listed Venture Capital	Venture Capital style investments into small listed companies with quality governance board & quality earnings poised to grow	Model-based	-	Lotusdew Wealth	PAID	Small cap	Low	NIFTY 500	14
	SCMO_0010	Value and Momentum	Positive momentum stocks available at a discount to their peers	Model-based	Q	Windmill Capital	PAID	Small cap	High	NIFTY 500	15
	WKIMO_0004	Mi_MT_Allcap	Pick Strongest Momentum among all highly liquid NSE stocks Past Performance>30%	Model-based	W	Weekend Investing	PAID	None	Low	NIFTY 500	15
	WKIMO_0007	Mi_ST_ATH	Chasing Strongest Momentum among All Time High stocks Past Performance>35% Weekly Rebalanced	Model-based	W	Weekend Investing	PAID	None	Low	NIFTY 500	10
MultiFactor	AGBMO_0012	ARQ Prime by Angel One	15 stock portfolio selected based on multi factor model and quantitative analytics	Model-based	-	Angel One	FREE	Multi cap	Medium	NIFTY 500	15
	SCSB_0003	Low Risk Smart Beta	A low volatile portfolio that aims for high returns. Passive investing with a twist	Smart Beta	Q	Windmill Capital	PAID	Large cap	Low	NIFTY 100	15
	TMHMO_0002	Tamohara Long Term Equity Strategy	Generating alpha by identifying pool of small and mid size companies.	Model-based	Q	Tamohara	PAID	None	Medium	NIFTY 500	14
	WRTMO_0003	Balanced Multi Factor	Tactical long term out-performance powered by multiple quantitative factors.	Model-based	M	Wright Research	PAID	None	Medium	NIFTY LARGEMID250	20-25
	WRTMO_0004	Growth Multi Factor	Tactical long term out-performance powered by multiple quantitative factors.	Model-based	M	Wright Research	PAID	None	Medium	NIFTY LARGEMID250	20-25
Quality	WRTMO_0005	Conservative Multi Factor	Tactical long term out-performance powered by multiple quantitative factors.	Model-based	M	Wright Research	PAID	None	Low	NIFTY LARGEMID250	20-25
	NIVMO_0001	Mid and Small Cap Focused Portfolio	This portfolio contains mid and small cap size companies available at reasonable valuations with strong market hold.	Model-based	-	Niveshaay	PAID	None	High	NIFTY 500	20
	QOVMO_0002	Qovar Quality	A portfolio that invests in large, growth oriented companies with strong fundamentals. Maintains relatively low portfolio risk.	Model-based	-	Qovar Advisors	PAID	None	Medium	NIFTY 100	15
	QOVMO_0003	Qovar Top 250 Growth	An actively managed portfolio that invests in large and medium size growth oriented companies.	Model-based	-	Qovar Advisors	PAID	None	Medium	NIFTY LARGEMID250	19
	QOVMO_0004	Qovar Top 500 Growth	A portfolio that invests in high-growth companies with sound fundamentals.	Model-based	-	Qovar Advisors	PAID	None	Medium	NIFTY 500	18
	SCMO_0007	Straight Flush	Quality stocks showing consistent earnings growth	Model-based	Q	Windmill Capital	PAID	Small cap	Medium	NIFTY 500	15
	SCMO_0024	The PE List	Companies in which PE and VC firms have high shareholding	Model-based	A	Windmill Capital	FREE	Small cap	Medium	NIFTY 500	15
	SCMO_0026	Growth and Income	Efficiently managed dividend paying companies screened using Kevin Matras' criteria	Model-based	Q	Windmill Capital	FREE	Small cap	Medium	NIFTY 500	15
	SCMO_0027	Zero Debt	Zero debt, high ROE companies with an established track record of high earnings growth	Model-based	A	Windmill Capital	FREE	Small cap	Medium	NIFTY 500	12
	SCMO_0029	CANSLIMesque	Efficiently managed growing companies experiencing positive momentum, screened using CANSLIM methodology	Model-based	Q	Windmill Capital	PAID	Multi cap	Medium	NIFTY 500	11
	SCNM_0007	Brand Value	Companies that own India's everyday brands. Top consumer favorites	Thematic	Q	Windmill Capital	PAID	Large cap	Medium	NIFTY 500	15
	SCSB_0001	Dividend Smart Beta	Companies increasing their dividends on a consistent basis	Smart Beta	Q	Windmill Capital	PAID	Large cap	Medium	NIFTY 100	15
	SCSB_0004	Quality Smart Beta	Established companies that have stood the test of time. Timeless classics	Smart Beta	A	Windmill Capital	PAID	Large cap	Medium	NIFTY 100	15
Value	CEDMO_0001	Sustainable Compounders	A portfolio of reasonably priced good quality equity shares and liquid ETF for sustainable returns	Model-based	-	Compound Everyday Capital	PAID	None	Medium	NIFTY 500	10-25
	MWNM_0003	Mystic Wealth Value (MWV)	MWV is a Value based portfolio sniffing inefficiencies in SpinOffs. management change and GARP	Thematic	Q	Mystic Wealth	PAID	None	Low	NIFTY 500	15
	SCMO_0003	Growth at a Fair Price	High growth companies that are available at fair valuations	Model-based	Q	Windmill Capital	PAID	Small cap	High	NIFTY 500	7
	SCMO_0016	The Naked Trader	Low-debt companies showing healthy growth. Criteria set by top trader Robbie Burns	Model-based	Q	Windmill Capital	FREE	Small cap	Medium	NIFTY 500	13

Source: Collated from Smallcase website and strategy fact sheets as on January 2023.

are not necessarily what the “Defining the Universe” section in the fact sheet says. For instance, “*All publicly traded companies on the National Stock Exchange of India, covering 90% market capitalization, are included in the universe*” for the smallcase CANSLIM-esque is categorised as Multicap and assigned Nifty 500 as the benchmark. In contrast, with the same universe, Dividend Aristocrats is classified as Largecap with the Nifty100 as its benchmark. On top of such differences, the dynamic nature of the process of assessing benchmarks poses a problem in evaluating the performance of a smallcase over time using a reference benchmark. The market cap

category in January 2023 could have differed from other points during our observation period. To overcome this problem of a reference benchmark changing, we use the universe of stocks that smallcase uses as the basis for the benchmark. By going through the fact sheets of all the smallcases in our sample, we have three benchmarks for our analysis: the Nifty 100, for smallcases which restrict their universe to approximate the top 100 stocks by market capitalisation, the Nifty Large and Mid Cap 250 for smallcases which restrict their universe to the top 300 stocks by market capitalisation, and the Nifty 500 index for larger universes. The benchmark we assign is shown in the column titled “Benchmark” in Table 1. Note that the benchmarks assigned can be different from what Smallcase would have shown on their platform.

The difference between the Smallcase and SEBI approaches is important for investors to be aware of. For example, Capitalmind Momentum is categorised as a Large Cap smallcase with an allocation of 38.0% to Large Cap firms, 23.5% to Mid Cap firms, and 3.5% to Small Cap firms⁶. As the sum of weights of the constituent large-cap stocks is greater than 50%, this smallcase is categorised as Largecap. However, the fact sheet states that the universe of stocks includes “*all publicly traded companies on the National Stock Exchange of India, covering 90% market capitalization*”. This is akin to the Nifty 500 universe, which would not be classified as a Large Cap. under SEBI’s definition of Large Cap schemes (At least 80% investment in large cap stocks⁷). Investors need to recognise that Smallcases Market Capitalisation classification, while using similar terminology to SEBI’s classification of mutual fund schemes, has a very different definition in many instances.

To show the “risk” of the inherent portfolios, Smallcase calculates⁸ the relative risk against the NIFTY 50 index for each strategy and categorises each smallcase into one of three risk categories: Low Volatility, Medium Volatility, and High Volatility. Smallcase warns investors that investing in “*High Volatility smallcases means that change in your investment values can be very sudden and drastic*”.

Table 1 also shows the number of holdings for each smallcase. The number of holdings in the smallcases in our sample is generally less than 20. Academic research has shown that portfolios with few holdings tend to have higher idiosyncratic risks (see, for instance, [Raju and Agarwalla \(2021\)](#)).

The Smallcase risk classification is based on a volatility ratio (VR), defined as “*Average annual rolling standard deviation of the smallcase divided by the average annual rolling standard deviation of the Nifty 50 Index*”. If $VR \geq 1.3$, the smallcase is categorised as High Volatility; if $0.8 \leq VR < 1.3$, it is classified as Medium Volatility, and if $VR < 0.8$, Low Volatility. In Table 2, we compute the annualised standard deviation for each smallcase in our sample using the three years of monthly returns. We add a 3-year VR using the annualised volatilities of the smallcase and the Nifty 50 TR Index using three years of month-end Net Asset Values. Using this 3-year VR, we use the same breakpoints used by Smallcase to calculate the 3-year Risk Category. It is

⁶As on January 2, 2023. The percentage of Large Cap exposure is $\frac{38}{38+23.5+3.5} = 58.5$

⁷Defined as stocks ranked between 1 and 100 by full market capitalisation.

⁸The calculation of relative risk is detailed in the fact sheet of each strategy.

clear from the table that the risk categories using our calculations are higher than those by Smallcase. Of the 36 smallcases in our sample, 10 funds were categorised as Low Volatility under the smallcase methodology. In contrast, only 2 smallcases are classified as Low using the three-year methodology. Of these two, *All Weather Investing*, is designed to be “*rebalanced periodically to generate relatively higher returns by assuming the least possible risks*”.

Table 2: Risk categorisation using Volatility Ratio calculated from 3-year monthly returns: December 2019 - December 2022

Style	code	Annual Std Dev	Smallcase Risk Cat	3yr Volatility Ratio	3yr Risk Category	Num Holdings
Asset Allocation	SCAW_0001	16.69	Low	0.75	Low	4
	SCET_0004	24.99	Medium	1.12	Medium	2
	SCET_0005	20.79	Low	0.93	Medium	2
	TRLMO_0003	21.07	Low	0.95	Medium	22
Dividends	SCMO_0013	30.78	Medium	1.38	High	11
	SCMO_0014	27.82	Medium	1.25	Medium	12
Low Vol	SCMO_0015	23.52	Medium	1.06	Medium	10
Momentum	CMMO_0001	27.22	Low	1.22	Medium	25
	FBCMO_0003	26.77	Medium	1.20	Medium	15
	FBCMO_0004	29.04	Medium	1.31	High	50
	LTDMO_0001	37.13	Low	1.67	High	14
	SCMO_0010	32.56	High	1.46	High	15
	WKIMO_0004	26.59	Low	1.20	Medium	15
	WKIMO_0007	28.08	Low	1.26	Medium	10
	AGBMO_0012	27.59	Medium	1.24	Medium	15
MultiFactor	SCSB_0003	22.76	Low	1.02	Medium	15
	TMHMO_0002	25.96	Medium	1.17	Medium	14
	WRTMO_0003	21.84	Medium	0.98	Medium	20-25
	WRTMO_0004	23.25	Medium	1.04	Medium	20-25
	WRTMO_0005	16.67	Low	0.75	Low	20-25
	NIVMO_0001	34.21	High	1.54	High	20
Quality	QOVMO_0002	25.93	Medium	1.17	Medium	15
	QOVMO_0003	26.78	Medium	1.20	Medium	19
	QOVMO_0004	29.09	Medium	1.31	High	18
	SCMO_0007	34.53	Medium	1.55	High	15
	SCMO_0024	29.39	Medium	1.32	High	15
	SCMO_0026	29.17	Medium	1.31	High	15
	SCMO_0027	28.48	Medium	1.28	Medium	12
	SCMO_0029	29.22	Medium	1.31	High	11
	SCNM_0007	26.33	Medium	1.18	Medium	15
	SCSB_0001	26.07	Medium	1.17	Medium	15
	SCSB_0004	25.33	Medium	1.14	Medium	15
	CEDMO_0001	22.70	Medium	1.02	Medium	10-25
	MWNM_0003	25.23	Low	1.13	Medium	15
	SCMO_0003	35.35	High	1.59	High	7
	SCMO_0016	32.96	Medium	1.48	High	13

Source: smallcases returns from www.smallcase.com, Nifty50 Total Return from Refinitiv

At the other end of the risk spectrum, the 3-year VR method has 12 smallcases compared to 3 strategies classified as High Volatility by Smallcase. The higher relative volatility of portfolios with a small number of holdings is consistent with the findings of several academic research⁹ reflecting the higher undiversified idiosyncratic risk of such portfolios. Investors should conduct their research and ensure that they are aware of the risks inherent in these portfolios.

3. Methodology and Data

3.1. Methodology

To examine factor exposures, we use three-factor models: the one-factor model (CAPM) ([Markowitz, 1952](#); [Sharpe, 1964](#)), the market factor, Fama-French and Carhart’s four-factor model ([Fama and French, 1993](#);

⁹See ?.

Carhart, 1997) adding Size, Value and Momentum factors, and the Fama-French five-factor model (Fama and French, 2015) further adding the Investment and Profitability factors plus momentum.

For the one-factor model, we regress the excess returns of each smallcase against the excess returns of its benchmark shown in Table 1. The resulting regression is as follows:

$$r_{i,t} - rf_t = \alpha + \beta(R_{b,t} - rf_t) + \epsilon \quad (1)$$

where $r_{i,t}$ is the return for the smallcase i in month t , rf_t is the risk-free rate for month t , and $R_{b,t}$ is the return for the benchmark b for month t . The alpha (α), the intercept term of the regression, is Jensen's alpha (Jensen, 1968), reflecting the outperformance of the smallcase relative to the benchmark. The alpha is a mixture of the manager's skill in the selection and allocation of stocks for the portfolio, market timing of entry and exits, and luck. The beta term (β) is the sensitivity of the smallcase to market risk. Nifty Indices calculates several strategy indices which "*are designed on the basis of quantitative models/investment strategies to provide a single value for the aggregate performance of a number of companies*"¹⁰. We show the regression results using Equation 1 between the smallcase and its appropriate Nifty strategy index as the benchmark. In doing so, we compare the smallcase returns against a comparable rule-based strategy. For instance, for a smallcase based on the momentum strategy using Nifty 300 as its universe, we would choose the Nifty 200 Momentum 30 Index¹¹ as the strategy benchmark. In contrast, for a smallcase with the Nifty 500 as the universe, we use the Nifty Midcap150 Momentum 50 Index¹².

The four-factor model is as follows.

$$r_{i,t} - rf_t = \alpha + \beta_{MKT}(R_{MKT,t} - rf_t) + \beta_{SMB}SMB_t + \beta_{HML}HML_t + \beta_{WML}WML_t + \epsilon \quad (2)$$

where $r_{i,t}$ is the return for the smallcase i in month t , rf_t is the risk-free rate, $R_{MKT,t}$ is the return for the market, and SMB_t , HML_t , and WML_t are the returns of the factors Size (Small-Minus-Big), Value (High-Minus-Low) and Momentum (Winner-Minus-Loser), respectively, for month t .

The six-factor model adds the Profitability and Investment factors to 2 as follows:

$$r_{i,t} - rf_t = \alpha + \beta_{MKT}(R_{MKT,t} - rf_t) + \beta_{SMB}SMB_t + \beta_{HML}HML_t + \beta_{RMW}RMW_t + \beta_{CMA}CMA_t + \beta_{WML}WML_t + \epsilon \quad (3)$$

where CMA_t and RMW_t are the returns for Profitability (Robust-Minus-Weak) and Investment (Conservative-Minus-Weak) factors for month t in addition to the variables described in 2.

¹⁰<https://www.niftyindices.com/indices/equity/strategy-indices>

¹¹See <https://www.niftyindices.com/indices/equity/strategy-indices/nifty200-momentum-30>

¹²See <https://www.niftyindices.com/indices/equity/strategy-indices/nifty-midcap150-momentum-50>. Note that we have not published the regression results for these smallcases against the alternative momentum index, which are weaker than the results using the Midcap Momentum Index.

In both Equations 2 and 3, the alpha term is a measure of outperformance relative to the respective factor model, and the beta terms show exposure to the specific factor.

Given the small number of holdings, most of the smallcases in our sample have a high undiversified idiosyncratic risk. So, *a priori*, we expect the goodness of fit for the regressions for 1, 2 and 3 to be moderate. We expect the goodness-of-fit to improve from the one-factor to the six-factor model. In addition to the regression of smallcases, we examine the factor exposures of the benchmark to better understand the output of 1, which uses different benchmarks. From Table 1, the benchmarks are based on size: from NIFTY 100 to NIFTY 500. As a result, the factor exposures for the benchmarks would intuitively show the exposure to the size factor (from negative for the NIFTY 100 to positive for the NIFTY 500), being the primary explainers of the benchmark returns.

We are interested in the coefficients of the exogenous variables for the regressions 2 and 3. Although we expect β_{SMB} to be clear evidence across styles, we hope to see statistically significant values for β_{WML} for smallcases categorised as Momentum, β_{HML} or β_{CMA} ¹³ for smallcases categorised under Value style. Generally, we expect the relevant factor coefficients to be statistically robust for the factor-based style categories (Momentum, Value, Quality). Smallcases are long-only portfolios, so there will be a lower expression of any factor in our regressions. However, we expect *a priori* to have statistically robust evidence. For multifactor and SmartBeta styles, the regression should result in multiple factors showing statistical evidence to explain the average returns of the smallcase. For smallcases categorised as Asset Allocation, we expect the market beta to be significantly lower than 1.

3.2. Data

The index value data for smallcases is taken from the Smallcase website¹⁴. The index values do not include transaction fees and other related costs. Smallcase highlights that no “*actual money was invested or trades were executed while calculating*” index values or smallcase performances. The index values are based on end-of-day prices of stocks/ETFs in a smallcase.

All Nifty benchmark total index information comes from Refinitiv or, if not, from the Nifty Indices website¹⁵. All index values for smallcases are in Indian rupees between December 2019 and December 2022. All values are month-end values, or failing that; the index values are on the date closest to the last trading day of the relevant month.

¹³Fama and French (2015) show that by adding RMW and CMA, for data covering the US markets (1963-2013), the HML factor is redundant since the time series of HML returns are entirely explained by the other four factors (most notably CMA).

¹⁴Smallcase’s methodology to calculate index values is available at <https://www.smallcase.com/meta/return-calculation>.

¹⁵Data available at <https://www.niftyindices.com/reports/historical-data>>Total Return Index Values

The factor data and risk-free rates are from Invespar’s “Data Library: Fama French 3 and 5 Factors and Momentum Factor for the Indian Market”¹⁶ (Raju, 2022b,a) until December 2022.

4. Results and Discussion

4.1. One Factor Analysis - Market Benchmarks

Table 3: Summary statistics for the one-factor model using Market Indices: Monthly return between December 2019 - December 2022

style	Code	Intercept		NIFTY 100		NIFTY 500		NIFTY LARGEMID250		R-squared	Ann Tracking Error
		Coeff	SE	Coeff	SE	Coeff	SE	Coeff	SE		
Asset Allocation	SCAW_0001	0.189	(0.311)	0.421***	(0.084)					0.57	15.33
	SCET_0004	0.206	(0.660)	0.883***	(0.132)					0.64	15.07
	SCET_0005	0.306	(0.484)	0.679***	(0.109)					0.64	13.41
	TRLMO_0003	0.429	(0.675)			0.618***	(0.127)			0.54	15.77
Dividends	SCMO_0013	1.056	(1.005)			1.012***	(0.193)			0.52	22.26
	SCMO_0014	1.000	(0.903)			0.882***	(0.128)			0.51	20.12
Low Vol	SCMO_0015	0.541	(0.661)			0.751***	(0.084)			0.57	15.88
Momentum	CMMO_0001	2.215**	(1.125)	0.870***	(0.150)	0.609***	(0.154)			0.26	25.45
	FBCMO_0003	0.059	(0.753)							0.52	19.01
	FBCMO_0004	1.310	(0.993)			0.875***	(0.176)			0.45	22.46
	LTDMO_0001	2.435**	(1.105)			1.244***	(0.207)			0.51	28.61
	SCMO_0010	1.845*	(1.105)			1.056***	(0.193)			0.49	24.48
	WKIMO_0004	2.361**	(0.996)			0.653***	(0.143)			0.31	23.68
	WKIMO_0007	2.514*	(1.300)			0.502***	(0.149)			0.16	28.72
MultiFactor	AGBMO_0012	1.645*	(0.873)	0.713***	(0.128)	0.863***	(0.125)			0.49	20.25
	SCSB_0003	0.159	(0.538)							0.54	16.02
	TMHMO_0002	0.357	(0.830)			0.792***	(0.152)			0.49	19.23
	WRTMO_0003	1.506*	(0.784)					0.517***	(0.078)	0.36	19.96
	WRTMO_0004	0.815	(0.748)					0.643***	(0.072)	0.47	18.44
	WRTMO_0005	0.567	(0.523)					0.299***	(0.059)	0.33	19.54
Quality	NIVMO_0001	2.234*	(1.174)	0.922***	(0.125)	1.129***	(0.196)			0.51	25.80
	QOVMO_0002	0.118	(0.673)					0.893***	(0.102)	0.63	15.86
	QOVMO_0003	-0.200	(0.705)							0.61	16.98
	QOVMO_0004	0.220	(0.953)			0.914***	(0.136)			0.49	21.62
	SCMO_0007	0.458	(1.122)			1.113***	(0.161)			0.48	26.70
	SCMO_0024	0.496	(0.897)			1.001***	(0.180)			0.57	19.92
	SCMO_0026	1.197	(0.849)			1.010***	(0.152)			0.59	19.28
	SCMO_0027	-0.417	(0.899)			0.898***	(0.159)			0.50	20.87
	SCMO_0029	0.826	(1.026)			0.871***	(0.188)			0.44	22.85
	SCNM_0007	0.162	(0.812)			0.849***	(0.154)			0.54	18.25
	SCSB_0001	-0.090	(0.628)	0.931***	(0.129)					0.64	15.86
	SCSB_0004	-0.078	(0.701)	0.786***	(0.131)					0.49	18.61
Value	CEDMO_0001	0.236	(0.689)			0.688***	(0.121)			0.53	16.42
	MWNM_0003	1.670**	(0.784)			0.696***	(0.119)			0.41	20.44
	SCMO_0003	0.832	(1.370)			0.968***	(0.210)			0.34	30.66
	SCMO_0016	1.100	(1.079)			1.061***	(0.168)			0.49	25.05

Source: smallcases returns from www.smallcase.com, Nifty Index Total Returns from Refinitiv, Nifty Indices, and risk-free rate from Data Library: Fama French 3 and 5 Factors and Momentum Factor for the Indian Market (* p<.1, ** p<.05, ***p<.01')

Table 3 shows the summary statistics of the one-factor regression for all the smallcases in our sample against the broad market indices. These regressions use the respective TR benchmark indices as the exogenous variable. As expected, given the concentrated portfolios, the goodness of fit, as shown by the R squared, is moderate: the median R-squared is 0.50 (minimum: 0.16 and maximum: 0.64.). The market-cap benchmark does not explain the average smallcases returns well. One cause for this moderate R-squared could be that the chosen benchmark is inappropriate. The Annualised Tracking Errors (ATE) for each smallcase against its chosen benchmark¹⁷ are consistently high, indicating that the concentrated portfolios have return profiles very different to their chosen benchmarks. For example, the smallcase with the highest ATE (30.7), Growth at a Fair Price, has only 7

¹⁶Available at <https://invespar.com/research>.

¹⁷Annualised Tracking Error = $\sqrt{\frac{12}{n-1} \sum_{t=1}^n (r_{i,t} - r_{b,t})^2}$ where $r_{i,t}$ and $r_{b,t}$ are the returns for smallcase i and its benchmark b respectively for month t and $n = 36$.

holdings and one of the lower values of R-squared (0.34). Similarly, the smallcase with the lowest goodness of fit (0.16), Growth at a Fair Price, also with a small number of holdings (10), has one of the highest ATE (28.7). The mean ATEs by benchmark are high: NIFTY 100: 16.1%, NIFTY LARGEMID250: 18.7%, NIFTY 500: 22.3%. Seemingly, the mean ATEs monotonically increase as the universe increases, which intuitively aligns with the thesis that small- and mid-cap stocks have higher volatility and, as a consequence, a larger number of stocks are required to reduce the idiosyncratic risk. The concentrated portfolios of smallcases with universes that allow them to select stocks from a larger pool of mid- and small-cap stocks will show higher ATEs. The lower goodness of fit and the higher ATEs are two sides of the same coin. With such high tracking errors, the relevance of a market-cap benchmark to provide a reference point seems moot.

In any event, 4 smallcases, Capitalmind Momentum (CMMO_0001), Listed Venture Capital (LTDMO_0001), Mi_MT_Allcap (WKIMO_0004), and Mystic Wealth Value (MWV) (MWNM_0003) evidence statistically significant intercept coefficients at 5%. The null hypothesis of a zero value for the intercept is not rejected for the remaining smallcases in our sample. Therefore, these funds would not show any evidence of alpha over their respective broad market benchmark. Clearly, many smallcases do not show proof of outperforming the benchmark for their universe of stocks despite the more concentrated portfolios and frequent rebalancing. Put another way, there is no evidence of skill in the selection, allocation, or market-timing of these strategies relative to a broad market index for the observation period. As our data of asset values for smallcases does not include transaction costs inherent in the frequent rebalancing, the potential slippage associated with trading in the less liquid small- and mid-cap stocks, fees charged, and the impact of taxes, realised returns would likely be lower.

4.2. One Factor Analysis - Strategy Benchmarks

Next, given the style of the smallcases in our sample, we look at their performance relative to the strategy indices published by Nifty as follows:

- **Dividends:** the Nifty Dividend Opportunities 50 Index, designed to provide exposure to high-yielding companies¹⁸ is used.
- **Low Volatility:** the Nifty Low Volatility 50 Index¹⁹ that tracks the performance of the least volatile securities listed on the NSE is used.
- **Momentum:** if the universe of stocks for the smallcase is the Nifty 100, the Nifty200 Momentum 30 Index²⁰ that aims to track the performance of 30 high momentum stocks across the large and midcap Nifty universe is used. Otherwise, the Nifty Midcap150 Momentum 50 Index²¹ that aims to track the performance of the top 50 companies within the Nifty Midcap 150 based on their momentum score is used.

¹⁸<https://www.niftyindices.com/indices/equity/strategy-indices/nifty-dividend-opportunities-50>

¹⁹<https://www.niftyindices.com/indices/equity/strategy-indices/nifty-low-volatility-50>

²⁰<https://www.niftyindices.com/indices/equity/strategy-indices/nifty200-momentum-30>

²¹<https://www.niftyindices.com/indices/equity/strategy-indices/nifty-midcap150-momentum-50>

- **MultiFactor**: the Nifty Alpha Quality Low-Volatility 30 Index²² which designed to reflect the performance of a portfolio based on a top combination of Alpha, Quality and Low Volatility is used. This index takes exposure to multiple factors.
- **Quality**: similar to Momentum, depending on the universe of stocks for the smallcase either the Nifty200 Quality 30 Index²³ or the Nifty Midcap150 Quality 50 Index²⁴ is used.
- **Value**: the The Nifty500 Value 50 Index²⁵ is used.

In each case, the total return index is used. We exclude the “Asset Allocation” category as there is no corresponding strategy index. Table 4 summarises the results. The median goodness of fit (0.47) is comparable to the median R-squared for the regressions against the market indices (0.49 excluding the “Asset Allocation” smallcases). Interestingly, none of the smallcases classified under Momentum evidence any statistically significant alpha (the intercept). All three smallcases showing evidence of alpha against market indices show no such evidence against momentum strategy indices. Against the strategy indices, only 2 smallcases, Balanced Multi Factor, and Mid and Small Cap Focused Portfolio show evidence of alpha at 5% confidence. More broadly, against strategy indices that are more transparent in their rules, hold a larger number of stocks, and have lower rebalancing frequencies, the more opaque, more concentrated, and more frequently rebalanced smallcases do not show evidence of outperformance during the three-year observation period.

Table 4: Summary statistics for the one-factor model using Strategy Indices: Monthly Returns between December 2019 - December 2022

style	Code	Intercept		NIFTY200 MOMENTUM 30		NIFTY200 QUALITY 30		NIFTY500 VALUE 50		NIFTY ALPHA QUALITY LOW VOLATILITY 30		NIFTY DIV OPFS 50		NIFTY LOW VOLATILITY 50		NIFTY MIDCAP150 MOMENTUM 50		R-squared
		Coeff	SE	Coeff	SE	Coeff	SE	Coeff	SE	Coeff	SE	Coeff	SE	Coeff	SE	Coeff	SE	
Dividends	SCMO_0013	1.06	(1.23)									1.07***	(0.34)					0.38
	SCMO_0014	0.95	(1.11)									0.97***	(0.21)					0.40
Low Vol	SCMO_0015	0.40	(0.67)											0.98***	(0.12)			0.55
Momentum	CMMO_0001	1.55	(0.98)													0.63***	(0.12)	0.34
	FBCMO_0003	-0.12	(0.92)	0.79***	(0.22)													0.41
	FBCMO_0004	0.57	(1.02)													0.81***	(0.18)	0.47
	LTDMO_0001	1.41	(1.19)													1.14***	(0.19)	0.52
	SCMO_0010	1.06	(1.26)													0.93***	(0.19)	0.47
	WKIMO_0004	1.93*	(1.07)													0.55***	(0.16)	0.27
MultiFactor	WKIMO_0007	2.05*	(1.23)													0.48***	(0.15)	0.18
	AGBMO_0012	1.80*	(1.02)															0.35
	SCSB_0003	0.08	(0.58)							0.96***	(0.28)							0.51
	TMHMO_0002	0.49	(1.01)							0.90***	(0.21)							0.35
	WRTMO_0003	1.73**	(0.85)							0.89***	(0.29)							0.22
	WRTMO_0004	1.11	(0.94)							0.55***	(0.14)							0.27
Quality	WRTMO_0005	0.69	(0.52)							0.67***	(0.19)							0.20
	NIVMO_0001	2.18**	(1.03)							0.32***	(0.09)							0.55
	QOVMO_0002	0.13	(0.87)			1.01***	(0.25)							1.22***	(0.22)			0.49
	QOVMO_0003	-0.04	(0.72)											0.95***	(0.12)			0.61
	QOVMO_0004	0.19	(0.91)											0.98***	(0.15)			0.52
	SCMO_0007	0.32	(0.85)											1.28***	(0.18)			0.59
	SCMO_0024	0.47	(0.86)											1.07***	(0.19)			0.61
	SCMO_0026	1.18	(0.81)											1.06***	(0.17)			0.61
	SCMO_0027	-0.51	(0.82)											1.01***	(0.13)			0.59
	SCMO_0029	0.76	(0.88)											0.97***	(0.19)			0.50
	SCNM_0007	0.15	(0.80)											0.89***	(0.16)			0.56
	SCSB_0001	-0.15	(0.72)			1.10***	(0.24)											0.57
Value	SCSB_0004	-0.15	(0.74)			0.95***	(0.21)											0.47
	CEBMO_0001	0.11	(0.89)					0.46***	(0.13)									0.42
	MWNM_0003	1.53	(0.96)					0.47***	(0.14)									0.33
	SCMO_0003	0.51	(1.54)					0.71***	(0.16)									0.34
	SCMO_0016	0.81	(1.29)					0.76***	(0.17)									0.44

Source: smallcases returns from www.smallcase.com, Nifty Index Total Returns from Refinitiv, Nifty Indices, and risk-free rate from Data Library: Fama French 3 and 5 Factors and Momentum Factor for the Indian Market (* p<.1, ** p<.05, ***p<.01')

²² <https://www.niftyindices.com/indices/equity/strategy-indices/nifty-alpha-quality-low-volatility-30>

²³ <https://www.niftyindices.com/indices/equity/strategy-indices/nifty200-quality-30>

²⁴ <https://www.niftyindices.com/indices/equity/strategy-indices/nifty-midcap150-quality-50>

²⁵ <https://www.niftyindices.com/indices/equity/strategy-indices/nifty500-value-50>

4.3. Factor Exposures - Nifty Indices

Before examining the factor exposures of the smallcases, we look at the factor exposures of the Nifty indices used in subsections 4.1 and 4.2 using Equations 2 and 3. Table 5 shows the exposures using the four-factor (Panel 5a) and six-factor (Panel 5b) respectively²⁶. The Nifty 100 and Nifty 500 market cap indices show clear evidence of SMB and SMB5 exposure. The Nifty 100 has a larger negative coefficient since it has the largest 100 firms by market cap. The Nifty 500 shows a smaller negative size exposure, aligned with the Data Library using more than 4.400 firms in constructing the factor dataset. There is no other factor that contributes to explaining average returns for the two market cap indices under the six-factor model.

The Nifty LargeMidcap 250 reflects the performance of a portfolio of 100 large-cap companies and 150 mid-cap companies listed on the NSE, where the aggregate weight of large-cap and mid-cap stocks is 50% each. This index shows exposure to expensive firms (negative HML, significant at 1%) and, to a lesser extent, exposure to “weaker” profitability firms and “loser” momentum firms (both at a 5% level). The size exposure is minimal, but that is due to the weighting schema neutralising the effects of the size factor.

Table 5: Summary statistics of the four-factor and six-factor exposures for Nifty Indices: Monthly Returns between December 2019 - December 2022

		NIFTY 100	NIFTY 500	NIFTY LARGE MID 250	NIFTY DIV OPPS 50	NIFTY LOW VOLATIL- ITY 50	NIFTY200 MOMEN- TUM 30	NIFTY MID- CAP150 MOMEN- TUM 50	NIFTY ALPHA QUALITY LOW VOLATIL- ITY 30	NIFTY200 QUALTY30	NIFTY M150 QLTY50	NIFTY500 VALUE 50
Intercept	Coeff	-0.23*	-0.24***	-0.13	0.10	-0.03	-0.47	0.32	-0.12	-0.01	-0.38	0.25
	SE	(0.13)	(0.09)	(0.10)	(0.20)	(0.29)	(0.39)	(0.45)	(0.26)	(0.33)	(0.35)	(0.37)
MF	Coeff	1.05***	1.04***	1.04***	0.84***	0.80***	0.98***	1.12***	0.87***	0.87***	0.90***	0.87***
	SE	(0.02)	(0.02)	(0.02)	(0.04)	(0.05)	(0.08)	(0.07)	(0.05)	(0.07)	(0.06)	(0.07)
SMB	Coeff	-0.22***	-0.09***	0.04	-0.21***	0.01	0.08	0.17	-0.06	-0.07	0.32***	-0.16
	SE	(0.03)	(0.02)	(0.03)	(0.07)	(0.07)	(0.08)	(0.12)	(0.07)	(0.10)	(0.09)	(0.10)
HML	Coeff	-0.09*	-0.08**	-0.12**	-0.02	-0.28***	-0.00	-0.27	-0.43***	-0.48***	-0.32***	1.09***
	SE	(0.05)	(0.04)	(0.05)	(0.10)	(0.10)	(0.15)	(0.19)	(0.13)	(0.17)	(0.12)	(0.19)
WML	Coeff	-0.04*	-0.05**	-0.09***	-0.01	0.05	0.37***	0.18**	0.15***	0.10*	-0.05	-0.33***
	SE	(0.02)	(0.02)	(0.02)	(0.04)	(0.05)	(0.08)	(0.08)	(0.05)	(0.05)	(0.06)	(0.11)
R-squared		0.99	0.99	0.99	0.92	0.91	0.86	0.88	0.88	0.83	0.93	0.93

(a) Four-factor exposures

		NIFTY 100	NIFTY 500	NIFTY LARGE MID 250	NIFTY DIV OPPS 50	NIFTY LOW VOLATIL- ITY 50	NIFTY200 MOMEN- TUM 30	NIFTY MID- CAP150 MOMEN- TUM 50	NIFTY ALPHA QUALITY LOW VOLATIL- ITY 30	NIFTY200 QUALTY30	NIFTY M150 QLTY50	NIFTY500 VALUE 50
Intercept	Coeff	-0.21	-0.22**	-0.06	-0.24	-0.39	-0.77**	0.57	-0.51**	-0.55*	-0.53*	0.23
	SE	(0.16)	(0.11)	(0.10)	(0.21)	(0.33)	(0.35)	(0.52)	(0.22)	(0.32)	(0.28)	(0.47)
MF	Coeff	1.03***	1.03***	1.03***	0.93***	0.82***	1.08***	1.11***	0.94***	0.97***	0.91***	0.95***
	SE	(0.03)	(0.02)	(0.03)	(0.03)	(0.06)	(0.07)	(0.09)	(0.05)	(0.06)	(0.05)	(0.07)
SMB5	Coeff	-0.21***	-0.08***	0.03	-0.15**	0.12	0.16**	0.11	0.02	0.04	0.38***	-0.15
	SE	(0.04)	(0.03)	(0.03)	(0.08)	(0.08)	(0.08)	(0.15)	(0.08)	(0.12)	(0.08)	(0.11)
HML	Coeff	0.02	-0.04	-0.16***	0.00	-0.20*	-0.19	-0.44**	-0.39***	-0.45***	-0.45***	0.88***
	SE	(0.07)	(0.05)	(0.06)	(0.08)	(0.10)	(0.17)	(0.21)	(0.12)	(0.15)	(0.13)	(0.18)
RMW	Coeff	0.06	0.01	-0.06**	0.15***	0.26***	-0.01	-0.24	0.22***	0.28***	0.01	-0.20*
	SE	(0.04)	(0.03)	(0.03)	(0.05)	(0.07)	(0.11)	(0.15)	(0.07)	(0.09)	(0.11)	(0.12)
CMA	Coeff	-0.03	-0.02	-0.06	0.44***	0.29***	0.39**	-0.20	0.42***	0.57***	0.09	0.20
	SE	(0.07)	(0.04)	(0.06)	(0.08)	(0.10)	(0.17)	(0.29)	(0.11)	(0.16)	(0.14)	(0.17)
WML	Coeff	-0.06*	-0.05	-0.06**	-0.07	-0.08	0.37***	0.29**	0.04	-0.04	-0.08	-0.22*
	SE	(0.04)	(0.03)	(0.03)	(0.05)	(0.07)	(0.09)	(0.12)	(0.06)	(0.08)	(0.09)	(0.12)
R-squared		0.98	0.99	0.99	0.95	0.94	0.88	0.89	0.92	0.89	0.93	0.94

(b) Six-factor exposures

Source: Nifty Index Total Returns from Refinitiv, Nifty Indices, and Factor returns and risk-free rate from Data Library: Fama French 3 and 5 Factors and Momentum Factor for the Indian Market

(* p<.1, ** p<.05, ***p<.01')

The strategy indices for individual factors, such as Value and Momentum, show statistically significant ex-

²⁶See [Raju and Krishnan \(2022\)](#) for a more detailed analysis of factor exposures of factor indices

posure to HML and WML, respectively. The MidCap Momentum index shows a weaker exposure to momentum. Intuitively, this is not surprising given that the short side of mid- and small-cap firms is an essential contributor to the momentum factor and the indices are long-only. As expected, the low-volatility index shows the smallest beta to market. The Alpha Quality Low-Volatility, the Dividend Opportunity and the Nifty 200 Quality Indices are exposed to “robust” and “conservative” firms, again as expected. Additionally, Alpha Quality Low Volatility and Nifty 200 Quality Indices are “expensive”, aligned with other studies on quality firms²⁷. The Nifty Midcap Quality Index shows little exposure to Profitability (RMW) or Investment (CMA).

The goodness of fit is high: median R-squared of 0.92 for the four-factor model and 0.94 for the six-factor model. Factor models explain a significant proportion of the returns. This result bolsters the conclusions in the sub-sections above that the smallcases, despite their concentrated portfolios and frequent rebalancing, do not show evidence of outperforming the market and strategy indices, which in turn show appropriate exposure to factors by their design.

4.4. *Four-Factor Exposures*

Next, we examine the four-factor exposures for the smallcases in our sample. Table 6 shows the summary results of the four-factor regressions using monthly returns between December 2019 and December 2022. The explanatory power of the four-factor model measured by the R-squared is higher than that of the one-factor model (median R-squared of 0.65 for the four-factor compared to 0.50 of the one-factor model using market indices).

The higher explanatory power of the four-factor model is accompanied by no smallcase showing of statistically significant evidence of alpha. The median market factor exposure for the one-factor model is 0.87, similar to that of the four-factor model (0.86). Therefore, the improvement in explanatory power is due to the other three factors. The single factor with the most statistically significant exposure is Size (SMB). 72% of the smallcases have statistically significant exposure to SMB at the 5% level. The positive coefficient indicates that the exposure of these smallcases is to *Small* firms²⁸. No smallcase shows statistical exposure to Value (HML), and only 4 firms show statistical evidence of exposure to Momentum (WML). The four-factor exposure indicates that average returns from smallcases are significantly driven by *Small* size: a reflexion of concentrated portfolios.

4.5. *Six-Factor Exposures*

The six-factor model further improves the explanatory power of the average returns of the smallcases in our sample between December 2019 and December 2022. Table 7 summarises the results of the regression using Equation 3. The median R-squared increases further to 0.68. Again, there is no statistical evidence of alpha, as seen from the standard errors of the intercept coefficients. The median market factor coefficient remains the

²⁷See Raju and Teli (2022b).

²⁸See Raju (2022b) for definitions and the methodology of factor construction.

Table 6: Summary statistics for the four-factor model: Monthly Returns between December 2019 - December 2022

style	Code	Intercept		MF		SMB		HML		WML		R-squared
		Coeff	SE	Coeff	SE	Coeff	SE	Coeff	SE	Coeff	SE	
Asset Allocation	SCAW_0001	-0.29	(0.29)	0.43***	(0.09)	0.07	(0.10)	0.12	(0.13)	0.11	(0.10)	0.64
	SCET_0004	-0.60	(0.52)	0.86***	(0.09)	0.26*	(0.15)	0.07	(0.22)	0.01	(0.12)	0.75
	SCET_0005	-0.36	(0.40)	0.67***	(0.11)	0.12	(0.13)	0.14	(0.18)	0.10	(0.12)	0.72
	TRLMO_0003	-0.17	(0.52)	0.54***	(0.08)	0.50***	(0.13)	-0.12	(0.20)	-0.06	(0.10)	0.71
Dividends	SCMO_0013	-0.11	(0.80)	0.97***	(0.14)	0.65***	(0.21)	-0.08	(0.34)	0.17	(0.22)	0.64
	SCMO_0014	0.44	(1.11)	0.93***	(0.13)	0.28	(0.19)	-0.27	(0.34)	0.03	(0.23)	0.55
Low Vol	SCMO_0015	-0.18	(0.55)	0.75***	(0.08)	0.44***	(0.14)	-0.21	(0.23)	0.10	(0.15)	0.68
Momentum	CMMO_0001	0.56	(0.86)	0.66***	(0.16)	0.67***	(0.22)	0.17	(0.28)	0.62**	(0.25)	0.53
	FBCMO_0003	-1.05*	(0.63)	0.94***	(0.10)	0.36*	(0.19)	-0.08	(0.31)	0.24	(0.16)	0.65
	FBCMO_0004	-0.23	(0.90)	0.90***	(0.10)	0.69***	(0.18)	0.07	(0.32)	0.42*	(0.21)	0.66
	LTDMO_0001	0.51	(0.82)	0.94***	(0.11)	1.05***	(0.26)	0.66	(0.42)	0.16	(0.19)	0.72
	SCMO_0010	0.07	(0.77)	0.92***	(0.18)	0.76***	(0.22)	0.52*	(0.27)	0.37	(0.26)	0.68
	WKIMO_0004	1.00	(1.01)	0.69***	(0.18)	0.44**	(0.21)	0.27	(0.43)	0.47*	(0.26)	0.48
	WKIMO_0007	1.73*	(0.95)	0.39***	(0.12)	0.90***	(0.25)	-0.38	(0.31)	-0.09	(0.22)	0.38
MultiFactor	AGBMO_0012	0.67	(0.91)	0.80***	(0.09)	0.55***	(0.20)	0.01	(0.28)	0.09	(0.20)	0.61
	SCSB_0003	-0.57	(0.50)	0.69***	(0.10)	0.35**	(0.15)	-0.09	(0.22)	0.04	(0.13)	0.64
	TMHMO_0002	-0.71	(0.51)	0.78***	(0.10)	0.64***	(0.17)	-0.15	(0.24)	0.17	(0.15)	0.67
	WRTMO_0003	1.04	(0.88)	0.43***	(0.11)	0.29	(0.19)	0.18	(0.28)	0.01	(0.14)	0.41
	WRTMO_0004	0.24	(0.79)	0.54***	(0.08)	0.36**	(0.18)	0.22	(0.28)	0.01	(0.13)	0.54
	WRTMO_0005	0.25	(0.59)	0.22***	(0.08)	0.10	(0.12)	0.32*	(0.17)	0.01	(0.09)	0.40
Quality	NIVMO_0001	0.20	(0.79)	1.11***	(0.13)	1.00***	(0.21)	0.10	(0.32)	0.50**	(0.22)	0.76
	QOVMO_0002	-0.91*	(0.53)	0.99***	(0.09)	0.25	(0.15)	0.00	(0.24)	0.21	(0.15)	0.75
	QOVMO_0003	-1.03	(0.70)	0.95***	(0.10)	0.42**	(0.17)	-0.09	(0.30)	0.23	(0.18)	0.69
	QOVMO_0004	-1.15	(0.89)	0.92***	(0.13)	0.54**	(0.23)	0.18	(0.38)	0.34**	(0.17)	0.64
	SCMO_0007	-0.72	(0.98)	0.84***	(0.17)	0.96***	(0.29)	0.09	(0.30)	-0.15	(0.37)	0.65
	SCMO_0024	-0.56	(0.67)	0.93***	(0.13)	0.64***	(0.19)	-0.06	(0.26)	0.07	(0.16)	0.70
	SCMO_0026	-0.11	(0.59)	0.87***	(0.10)	0.63***	(0.18)	0.34	(0.24)	0.14	(0.18)	0.75
	SCMO_0027	-1.63**	(0.72)	0.87***	(0.13)	0.59***	(0.19)	0.05	(0.28)	0.26	(0.21)	0.62
	SCMO_0029	-0.75	(0.80)	0.86***	(0.09)	0.81***	(0.20)	0.01	(0.31)	0.39***	(0.14)	0.67
	SCNM_0007	-0.37	(0.67)	0.88***	(0.14)	0.44**	(0.18)	-0.45	(0.33)	-0.03	(0.13)	0.63
	SCSB_0001	-1.01*	(0.56)	1.00***	(0.10)	0.25	(0.17)	-0.12	(0.24)	0.17	(0.12)	0.73
	SCSB_0004	-1.22*	(0.66)	0.88***	(0.10)	0.33*	(0.18)	-0.06	(0.25)	0.33*	(0.17)	0.64
Value	CEDMO_0001	-0.66	(0.50)	0.64***	(0.10)	0.51***	(0.13)	0.01	(0.19)	0.12	(0.15)	0.69
	MWNM_0003	0.55	(0.70)	0.61***	(0.12)	0.63***	(0.20)	0.14	(0.29)	0.16	(0.17)	0.59
	SCMO_0003	-1.16	(0.97)	0.76***	(0.22)	1.05***	(0.27)	0.56	(0.38)	0.32	(0.33)	0.59
	SCMO_0016	-0.72	(0.72)	0.94***	(0.17)	0.90***	(0.20)	0.34	(0.29)	0.34	(0.27)	0.71

Source: smallcases returns from www.smallcase.com, monthly Factor returns and risk-free rate from Data Library: Fama French 3 and 5 Factors and Momentum Factor for the Indian Market
 (* p<.1, ** p<.05, ***p<.01')

same (0.80), which implies that additional factors increase the explanatory power.

The median exposure to SMB for the six-factor model (0.80) is lower than the exposure in the four-factor model (0.86). A smaller 64% of smallcases show statistically significant exposure to *Small Size*. For example, Qovar Top 500 Growth (QOVMO_0004), which showed a statistically significant exposure to SMB in the four-factor model (5% confidence), no longer shows this exposure in the six-factor model. Instead of SMB, the six-factor model shows statistically significant positive exposure to WML (at 1% confidence), with R-squared at 0.65 (compared to 0.64 for the four-factor model). The difference in the methodology for computing SMB in the four- and six-factor models (see [Fama and French \(2015\)](#); [Raju \(2022b\)](#)) causes subtle changes in factor exposure. No smallcase in our sample shows exposure to the *Big Size* factor during our observation period.

Within the styles, there are some interesting nuances. Asset Allocation smallcases tend to have low betas of market factor, except for Top 100 Stocks (SCET_004), which by design “*invests in top 100 marketcap companies*”. From Table 3, this smallcase has a high beta of Nifty 100, but the R-squared was only 0.64 and showed no evidence of alpha in the market index. The smallcase manager’s algorithm did not lead to superior performance relative to the Nifty 100 despite the high ATE. The four-factor regression shows a weakly significant (10% level) exposure to *BigSize*. For this smallcase, the evidence suggests that while the manager’s algorithm has a high tracking error to the Nifty 100, during our observation period, there is no evidence of outperformance over the

Table 7: Summary statistics for the six-factor model: Monthly Returns between December 2019 - December 2022

style	Code	Intercept		MF		SMB5		HML		RMW		CMA		WML		R-squared
		Coeff	SE	Coeff	SE	Coeff	SE	Coeff	SE	Coeff	SE	Coeff	SE	Coeff	SE	
Asset Allocation	SCAW_0001	-0.12	(0.38)	0.33***	(0.06)	0.07	(0.11)	0.23	(0.16)	0.04	(0.10)	-0.32*	(0.18)	0.08	(0.11)	0.68
	SCET_0004	-0.50	(0.61)	0.84***	(0.12)	0.23	(0.16)	-0.03	(0.29)	-0.10	(0.15)	-0.13	(0.27)	0.05	(0.11)	0.75
	SCET_0005	-0.09	(0.48)	0.59***	(0.09)	0.07	(0.14)	0.20	(0.22)	-0.06	(0.12)	-0.38*	(0.23)	0.12	(0.13)	0.73
	TRLMO_0003	-0.03	(0.56)	0.45***	(0.08)	0.49***	(0.15)	-0.16	(0.22)	-0.02	(0.10)	-0.32	(0.23)	-0.07	(0.09)	0.73
Dividends	SCMO_0013	-0.14	(0.92)	0.89***	(0.14)	0.73***	(0.21)	-0.20	(0.38)	0.01	(0.20)	-0.23	(0.48)	0.13	(0.23)	0.65
	SCMO_0014	0.61	(1.16)	0.90***	(0.17)	0.23	(0.22)	-0.37	(0.44)	-0.13	(0.30)	-0.21	(0.41)	0.09	(0.29)	0.56
Low Vol	SCMO_0015	-0.51	(0.63)	0.81***	(0.11)	0.51***	(0.16)	-0.37	(0.29)	0.08	(0.12)	0.29	(0.32)	0.04	(0.15)	0.68
Momentum	CMMO_0001	0.54	(1.05)	0.61***	(0.14)	0.74***	(0.25)	-0.03	(0.34)	-0.07	(0.19)	-0.18	(0.39)	0.61**	(0.26)	0.55
	FBCMO_0003	-1.14	(0.76)	0.93***	(0.15)	0.42**	(0.21)	-0.21	(0.41)	-0.01	(0.18)	0.00	(0.34)	0.23*	(0.14)	0.66
	FBCMO_0004	-0.20	(1.16)	0.91***	(0.14)	0.69***	(0.21)	-0.24	(0.38)	-0.18	(0.26)	-0.07	(0.45)	0.48**	(0.22)	0.66
	LTDMO_0001	0.91	(0.93)	0.87***	(0.15)	1.02***	(0.26)	0.11	(0.44)	-0.52***	(0.17)	-0.52	(0.41)	0.36*	(0.19)	0.76
	SCMO_0010	0.67	(1.03)	0.70***	(0.14)	0.69***	(0.26)	0.47*	(0.28)	-0.20	(0.16)	-0.95**	(0.44)	0.44*	(0.26)	0.72
	WKIMO_0004	0.87	(1.14)	0.69***	(0.22)	0.47*	(0.26)	0.21	(0.54)	0.07	(0.34)	0.02	(0.47)	0.42	(0.28)	0.48
	WKIMO_0007	1.54	(0.98)	0.42***	(0.15)	0.96***	(0.28)	-0.78*	(0.42)	-0.13	(0.30)	0.10	(0.47)	-0.07	(0.25)	0.39
MultiFactor	AGBMO_0012	0.62	(1.10)	0.84***	(0.14)	0.59**	(0.24)	-0.36	(0.30)	-0.22	(0.18)	0.06	(0.41)	0.17	(0.21)	0.62
	FCSB_0003	-0.93*	(0.55)	0.66***	(0.11)	0.47***	(0.14)	-0.01	(0.28)	0.31*	(0.16)	0.14	(0.28)	-0.13	(0.15)	0.68
	TMHMO_0002	-0.58	(0.51)	0.75***	(0.14)	0.63***	(0.17)	-0.42	(0.32)	-0.21	(0.16)	-0.20	(0.36)	0.25	(0.18)	0.68
	WRTMO_0003	0.60	(0.96)	0.59***	(0.13)	0.40*	(0.21)	-0.21	(0.35)	-0.10	(0.21)	0.59**	(0.30)	0.04	(0.14)	0.46
	WRTMO_0004	-0.00	(0.90)	0.65***	(0.12)	0.40*	(0.21)	-0.13	(0.35)	-0.15	(0.18)	0.36	(0.28)	0.07	(0.11)	0.56
	WRTMO_0005	-0.09	(0.60)	0.30***	(0.09)	0.19	(0.12)	0.19	(0.22)	0.06	(0.13)	0.40**	(0.19)	-0.02	(0.09)	0.44
Quality	NIVMO_0001	-0.26	(0.98)	1.30***	(0.17)	1.06***	(0.25)	-0.56	(0.40)	-0.23	(0.22)	0.61	(0.40)	0.57***	(0.20)	0.77
	QOVMO_0002	-0.80	(0.64)	0.96***	(0.13)	0.23	(0.17)	-0.09	(0.31)	-0.09	(0.13)	-0.15	(0.28)	0.24*	(0.13)	0.75
	QOVMO_0003	-0.92	(0.91)	0.95***	(0.14)	0.39*	(0.21)	-0.32	(0.36)	-0.19	(0.18)	-0.12	(0.34)	0.31*	(0.16)	0.70
	QOVMO_0004	-0.95	(1.14)	0.94***	(0.17)	0.47*	(0.27)	-0.18	(0.45)	-0.34	(0.21)	-0.14	(0.38)	0.49***	(0.16)	0.65
	SCMO_0007	-1.08	(1.01)	0.80***	(0.19)	1.12***	(0.30)	-0.11	(0.35)	0.15	(0.20)	0.05	(0.49)	-0.28	(0.39)	0.67
	SCMO_0024	-0.74	(0.67)	0.86***	(0.13)	0.73***	(0.19)	-0.08	(0.29)	0.18	(0.18)	-0.10	(0.32)	-0.05	(0.16)	0.71
	SCMO_0026	0.03	(0.72)	0.84***	(0.16)	0.61***	(0.21)	0.10	(0.32)	-0.18	(0.15)	-0.23	(0.34)	0.20	(0.19)	0.75
	SCMO_0027	-1.41*	(0.79)	0.80***	(0.16)	0.58***	(0.19)	-0.19	(0.37)	-0.22	(0.22)	-0.35	(0.37)	0.33	(0.22)	0.65
	SCMO_0029	-0.22	(1.15)	0.72***	(0.15)	0.73***	(0.25)	-0.21	(0.31)	-0.33**	(0.17)	-0.75*	(0.45)	0.52***	(0.18)	0.71
	SCNM_0007	0.15	(0.71)	0.80***	(0.19)	0.28	(0.20)	-0.61	(0.43)	-0.33	(0.20)	-0.55	(0.42)	0.12	(0.16)	0.65
	FCSB_0001	-1.07	(0.66)	0.94***	(0.14)	0.29	(0.20)	-0.02	(0.31)	0.17	(0.16)	-0.12	(0.31)	0.07	(0.13)	0.74
	FCSB_0004	-1.57**	(0.76)	0.88***	(0.13)	0.44**	(0.21)	-0.02	(0.30)	0.26*	(0.15)	0.19	(0.35)	0.18	(0.16)	0.66
Value	CEDMO_0001	-0.38	(0.62)	0.57***	(0.11)	0.46***	(0.14)	-0.15	(0.24)	-0.20	(0.14)	-0.38	(0.29)	0.20	(0.16)	0.71
	MWNM_0003	0.32	(0.89)	0.61***	(0.16)	0.70***	(0.23)	-0.03	(0.40)	0.06	(0.20)	0.10	(0.32)	0.10	(0.16)	0.59
	SCMO_0003	-1.20	(1.29)	0.78***	(0.22)	1.08***	(0.34)	0.06	(0.48)	-0.27	(0.29)	-0.03	(0.66)	0.41	(0.36)	0.60
	SCMO_0016	-1.13	(0.78)	1.04***	(0.20)	1.03***	(0.21)	-0.16	(0.37)	-0.11	(0.16)	0.39	(0.42)	0.35	(0.25)	0.72

Source: smallcases returns from www.smallcase.com, monthly Factor returns and risk-free rate from Data Library: Fama French 3 and 5 Factors and Momentum Factor for the Indian Market

(* p<.1, ** p<.05, ***p<.01')

large-cap index.

TrendlineZ Balanced (TRLMO_0003) is another interesting smallcase under Asset Allocation. It has a low beta of the market factor and a statistically significant (1% level) bias toward *Small* Size. This exposure is also seen in the four-factor regression (Table 6). During the observation period, the manager has delivered a low beta strategy with a concentrated 22-stock portfolio mainly exposed to *Small* Size, reflecting the skill in stock selection, allocation, and market timing. However, this skill does not translate into alpha over the Nifty 500 (Table 4), rather, better risk-adjusted returns. Like other smallcases exposed to the *Small* Size factor, the market microstructure implies that these strategies will quickly reach capacity given the low liquidity of mid- and small-cap stocks that make up such portfolios.

The two Dividends smallcases, Dividend Stars (SCMO_0013) and Dividend Aristocrats (SCMO_0014) differ in their exposure to Size, with the former exposed to *Small* Size to a larger degree (also seen in Table 6). The manager shows high tracking errors to the underlying universe and the Nifty Dividend strategy ETF (Tables 3 and 4) with a concentrated portfolio (11/12 stocks). Dividend Stars, given exposure to *Small* Size, has a higher realised volatility (Table 2) than Dividend Aristocrats. Unfortunately, in absolute return terms, there is no evidence of alpha during the period.

Among the smallcases of Momentum, 2 smallcases, Capitalmind Momentum (CMMO_0001), and Flameback MultiCap Momentum Factor (FBCMO_0004), show statistically significant (5% level) evidence of exposure to Momentum factor (WML). Both are exposed to *Small* Size and show betas of the market factor lower than

those of the two indices of the Nifty Momentum strategy (Table 5b) with higher WML coefficients. Both show positive alpha values versus the strategy benchmark, but the coefficients are not significant (Table 4). Capitalmind Momentum holds 25 stocks while Flameback Multi-Cap Momentum Factor holds 50 stocks, the largest in our smallcase sample (Table 1). While the managers' evidence skill in managing these two smallcases, given the *Small* Size exposure, will capacity limit the ability to scale? The other smallcases under this category do not show statistical evidence of exposure to the Momentum factor - possibly due to the high idiosyncratic risks arising from the concentrated portfolios.

Among the small cases of Multifactor, 2 smallcases, Balanced Multi Factor (WRTMO_0003), and Conservative Multi Factor (WRTMO_0005), show statistically significant (5% level) evidence of exposure to Investment factor (CMA) while not showing statistically significant exposure to *Small* Size. There is no evidence of exposure to the Value factor claimed by the manager, and the low market factor beta is evidence of a low volatility tilt of the strategy. The Conservative Multi-Factor is less volatile than the Balanced Multi-Factor, and Conservative Multi-Factor (see also Table 2). Against our chosen Multifactor strategy benchmark (Nifty Alpha Quality Low Volatility 30 Index, the Balanced Multi-Factor does evidence statistically significant alpha before fees. However, alpha is not evident against their market index, with low goodness of fit for the regression.

In other styles, Quality and Value, there is no evidence to show exposure to academic Value or Profitability and Investment factors associated with Quality. As seen with several smallcases, the results are likely due to the idiosyncratic risk associated with concentrated portfolios, more than overwhelming any factor exposure the manager aims to take. Despite frequent rebalancing, the size exposure seems to be consistently evidenced. The high idiosyncratic risk also makes explaining average returns by factor models difficult - hence the lower goodness of fits. The six-factor model is the best among the various factor models but only slightly better than the four-factor model. Despite the methodological differences in the construction of the Size factor (SMB) between the four and six-factor models, exposure to *Small* Size remains in clear evidence. The strategy benchmarks, with larger portfolios, evidence factor tilts more clearly. However, even these have varying exposure to the factors returns they aim to track. Investors in smallcases should consider the effects of idiosyncratic risk in concentrated portfolios. Our results show evidence of manager skill, but realised returns after costs and taxes must be accounted for before deciding to invest.

The *Small* bias is likely to increase operational costs in the form of slippage and the loss of opportunity of low liquidity. Market liquidity for small- and mid-cap stocks is also expected to constrain the capacity for smallcase strategies. With numerous strategies (more than 1,000, as reported on the Smallcase website), the capacity constraint in aggregate demand for a stock is genuine. For many stocks in the bottom 200 of the Nifty 500 constituents, the volume traded is likely to be less than a few crores on any day. If the smallcase managers use similar algorithms, the same stocks will be picked by algorithms making for high slippage costs and significantly lower realised returns for investors.

5. Conclusion

The Smallcase Technologies claims to build “platforms & investment products to invest better in Indian equities”. They define a smallcase as a *basket of stocks/ETFs curated to reflect an idea*. Several smallcases are systematic rule-based concentrated portfolios that draw reference from academic factors. Many make specific references to well-regarded international academic studies in their material. In doing so, their proprietary algorithms, similar to factor strategy tilt indices, pick stocks demonstrating high levels of a particular factor characteristic across the cross section of their respective universe of stocks. While factor strategy indices generally have a larger number of holdings at any point in time and are fully transparent in their methodologies, smallcase managers are more opaque in their methodology, except for general reference to their approach, stay agile in their rebalancing, and hold concentrated portfolios. This paper examines the returns of 36 smallcases between December 2019 and December 2022 to determine their factor exposure, comparing the performance against market-based indices, strategy indices, and four- and six-factor models.

Despite the small sample (in terms of the number of small cases and the 3-year observation period) due to the newness of Smallcase and its business, there are some preliminary conclusions that the evidence shows. First, some smallcases show clear evidence of exposure to the targeted factor. Second, most small cases are exposed to the *Small Size* factor. Concentrated portfolios with high idiosyncratic risk associated with mid- and small-cap stocks hide any exposure to the targeted factor when standard techniques like return-based factor models are used. Third, evidence of academic alpha is also not seen. The data we use are gross fees, slippage, transaction costs, and taxes. Realisable returns are likely to be lower²⁹. A further consequence of this exposure to the *Small Size* factor is that these strategies will likely face capacity constraints that will further depress realisable returns.

Finally, while the terminology for categorisation used on the Smallcase platform is similar to that adopted in the more tightly regulated fund management industry, the actual calculations used to determine the category differ. For example, “LargeCap” funds under SEBI’s classification schema for equity mutual funds are different from the “LargeCap” market cap category adopted by Smallcase.

The Smallcase story is one of innovation in the wealth industry, with new value chains being created for brokers, SEBI registered asset managers and investment professionals, wealth managers, technology providers, and investors. Smallcase also allows non-licensed/quasi-professional stock market enthusiasts to build strategies they could recommend to their friends and family³⁰. This innovation is still in its early phase and, as it grows, it will likely be of interest to more research as well as regulatory standardisation with the fund management industry.

²⁹See Raju and Teli (2022a) for a method to estimate costs of systematic factor strategies.

³⁰See “How Amazon-backed investment startup smallcase grew its userbase 7X in 3 years, March 4, 2022, at <https://yourstory.com/2022/03/amazon-backed-investment-platform-smallcase-grows-7x-user-base>.

References

- Carhart, M. M., 1997. On Persistence in Mutual Fund Performance. *The Journal of Finance* 52, 57–82.
- Fama, E. F., French, K. R., 1993. Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics* 33, 3–56.
- Fama, E. F., French, K. R., 2015. A Five-Factor Asset Pricing Model. *Journal of Financial Economics* 116, 1–22.
- Jensen, M. C., 1968. The Performance of Mutual Funds in the Period 1945–1964. *The Journal of Finance* 23, 389–416.
- Markowitz, H., 1952. Portfolio selection. *The Journal of Finance* 7, 77–91.
- Raju, R., 2022a. A Five-Factor Asset Pricing Model: Preliminary Evidence from India. Available at SSRN 4190426 p. 33.
- Raju, R., 2022b. Four and Five-Factor Models in the Indian Equities Market. SSRN eLibrary p. 37.
- Raju, R., Agarwalla, S. K., 2021. Equity portfolio diversification: how many stocks are enough? Evidence from India. IIMA Working Papers WP 2021-02-02, Indian Institute of Management Ahmedabad, Research and Publication Department.
- Raju, R., Krishnan, H., 2022. Factor Indices in India: Factor Exposures and Style Analysis. Available at SSRN p. 26.
- Raju, R., Teli, A., 2022a. 'Long' Factors, not 'Short' Change: Long Only Factor Portfolios in India. Available at SSRN 4000418 .
- Raju, R., Teli, A., 2022b. The Conservative Formula: Evidence from India. SSRN eJournal p. 23.
- Sharpe, W. F., 1964. Capital Asset Prices: A Theory of Market Equilibrium Under Conditions of Risk. *The Journal of Finance* 19, 425–442.